University of Copenhagen
Faculty of Science
Department of Chemistry

Optimized Virtual Orbital Space for Quantum Computing

Master's Thesis in Chemistry / Quantum Chemistry

Author
Tobias ⟨*Surname*⟩
xxx@ku.dk

Supervisor
⟨Supervisor Name⟩
Department of Chemistry, UCPH

Co-supervisor (if any)
⟨Co-supervisor Name⟩

Copenhagen, March 25, 2026

Abstract

⟨*Write the abstract here after all chapters are complete.*⟩

Keywords: Optimized virtual orbital space, OVOS, MP2, orbital optimisation, virtual orbital reduction, quantum computing, PySCF, Newton-Raphson.

Resumé

⟨*Danish translation of the abstract.*⟩

Acknowledgements

⟨*Write acknowledgements here.*⟩

List of Abbreviations

HF	Hartree-Fock
RHF	Restricted Hartree-Fock
UHF	Unrestricted Hartree-Fock
MP2	Second-Order Møller-Plesset Perturbation Theory
MPn	Møller-Plesset Perturbation Theory of order n
CI	Configuration Interaction
CISD	Configuration Interaction Singles and Doubles
FCI	Full Configuration Interaction
CC	Coupled Cluster
CCSD	Coupled Cluster Singles and Doubles
MCSCF	Multi-Configurational Self-Consistent Field
CASSCF	Complete Active Space Self-Consistent Field
DFT	Density Functional Theory
OVOS	Optimized Virtual Orbital Space
FNO	Frozen Natural Orbitals
PNO	Pair Natural Orbitals
DLPNO	Domain-Based Local Pair Natural Orbitals
NO	Natural Orbitals
VQE	Variational Quantum Eigensolver
QPE	Quantum Phase Estimation
JW	Jordan-Wigner
BK	Bravyi-Kitaev
NISQ	Noisy Intermediate-Scale Quantum

QC	Quantum Computing
BO	Born-Oppenheimer
AO	Atomic Orbital
MO	Molecular Orbital
SCF	Self-Consistent Field
LCAO	Linear Combination of Atomic Orbitals
ERI	Electron Repulsion Integral
CBS	Complete Basis Set
GTO	Gaussian-Type Orbital
STO	Slater-Type Orbital
DIIS	Direct Inversion in the Iterative Subspace
NR	Newton-Raphson
RLE	Reduced Linear Equation
PySCF	Python-based Simulations of Chemistry Framework
UCPH	University of Copenhagen

Contents

Abstract iii

Resumé v

Acknowledgements vii

List of Abbreviations ix

1 Introduction 1
- 1.1 Motivation . . . 1
- 1.2 The Virtual Orbital Space Problem . . . 1
- 1.3 Quantum Computing Context . . . 1
- 1.4 Objectives . . . 1
- 1.5 Thesis Outline . . . 1

2 Background 3
- 2.1 The Electronic Structure Problem . . . 3
- 2.2 Hartree-Fock Theory . . . 3
  - 2.2.1 Restricted and Unrestricted Formalism . . . 3
- 2.3 Molecular Orbital Integrals . . . 3
  - 2.3.1 One-Electron Integrals . . . 3
  - 2.3.2 Two-Electron Integrals . . . 3
  - 2.3.3 Integral Notation: Chemists' vs. Physicists' Conventions . . . 3
- 2.4 Møller-Plesset Perturbation Theory . . . 3
  - 2.4.1 MP2 Correlation Energy . . . 3
  - 2.4.2 Spin-Orbital Formulation . . . 3
- 2.5 Spin-Orbital Formalism . . . 3
- 2.6 Orbital Space Partitioning and Index Notation . . . 3
- 2.7 Brueckner Orbitals . . . 3
  - 2.7.1 Definition and Physical Interpretation . . . 3
  - 2.7.2 Brueckner Orbitals in Coupled Cluster Theory . . . 3
  - 2.7.3 Connection to Orbital-Optimised MP2 . . . 3
- 2.8 Quantum Computing and Orbital Space . . . 4
  - 2.8.1 Second Quantisation and Qubit Mapping . . . 4
  - 2.8.2 The Variational Quantum Eigensolver . . . 4
  - 2.8.3 OVOS Orbitals as a VQE Reference State . . . 4

3 Theory **5**
3.1 The Second-Order Hylleraas Functional 5
3.2 Orbital Rotation Parameterisation 5
3.2.1 Active and Inactive Virtual Subspaces 5
3.2.2 Unitary Transformation via Matrix Exponential 5
3.3 Gradient of J_2 with Respect to Orbital Rotations 5
3.3.1 MP2 Virtual-Virtual Density Matrix 5
3.3.2 Gradient Expression 5
3.4 Hessian of J_2 with Respect to Orbital Rotations 5
3.4.1 Block Structure of the Hessian 5
3.4.2 Negative Eigenvalues and Level-Shift Stabilisation 5
3.5 Newton-Raphson Orbital Optimisation 5
3.6 Generating the Unitary Rotation Matrix 5
3.7 Orbital Canonicalisation 5
3.8 Energy Recovery and the OVOS Error Metric 5
3.9 OVOS Orbitals as Approximate Brueckner Orbitals 6
3.9.1 Orbital Stationarity and the Brueckner Condition 6
3.9.2 Relationship to Orbital-Optimised MP2 6
3.9.3 Expected Reduction of T1 Amplitudes 6
3.10 OVOS Molecular Orbitals as a Trial State for VQE 6
3.10.1 Reference-State Overlap with the FCI Ground State 6
3.10.2 Reduction of T1 Amplitudes and Circuit Depth 6
3.10.3 Qubit Resource Savings 6

4 Method **7**
4.1 Algorithm Overview 7
4.2 Integral Transformation 7
4.2.1 AO-to-MO Transformation 7
4.2.2 Spin-Orbital Construction 7
4.2.3 Notation Convention: Chemists' vs. Physicists' Integrals 7
4.3 Initial Active Space Selection 7
4.4 MP1 Amplitudes and MP2 Energy 7
4.5 Gradient and Hessian Assembly 8
4.5.1 Block Structure and Efficient Inversion 8
4.5.2 Level-Shift Stabilisation 8
4.6 Orbital Rotation and Update 8
4.7 Convergence and Iteration Control 8
4.8 Initialisation Strategies 8
4.8.1 Default SCF Initialisation 8
4.8.2 Warm-Start from Previous Orbital Set 8
4.8.3 Random Unitary Sampling 8

5 Computational Details **9**
5.1 Software and Libraries 9
5.2 Hardware 10
5.3 Molecular Geometries 10
5.4 Basis Sets 10
5.5 Reference SCF Calculations 10

5.6 OVOS Calculation Settings . . . 10
5.7 Data Storage and Reproducibility . . . 10

6 Results **11**
6.1 Validation of the Implementation . . . 11
6.2 Energy Recovery as a Function of Active Virtual Orbital Count . . . 12
6.3 Convergence Behaviour . . . 14
6.4 Effect of Initialization Strategy . . . 15
6.5 Basis Set Dependence . . . 16
6.6 Comparison with Adamowicz and Bartlett (1987) . . . 17
6.7 Computational Cost . . . 18
6.8 T1 Amplitudes and the Brueckner Orbital Connection . . . 18
6.9 OVOS Orbitals as a VQE Reference State: Resource Estimates . . . 20

7 Conclusion **21**
7.1 Summary of Findings . . . 21
7.2 Critical Assessment . . . 21
7.3 Implications for Quantum Computing . . . 21

8 Outlook **23**
8.1 Algorithmic Improvements . . . 23
8.2 Method Extensions . . . 23
8.3 Quantum Computing Applications . . . 23
8.3.1 OVOS Orbitals as a VQE Reference State . . . 23
8.3.2 Brueckner Orbitals and Circuit Depth on NISQ Devices . . . 23
8.3.3 Qubit Count Reduction and Fault-Tolerant Perspectives . . . 23
8.4 Larger Molecular Systems . . . 23
8.5 Benchmarking and Validation . . . 23

Bibliography **25**

A Mathematical Derivations **27**
A.1 Spin-Orbital MP2 Energy from the Hylleraas Functional . . . 27
A.2 Derivation of the Orbital Rotation Gradient . . . 27
A.3 Derivation of the Orbital Rotation Hessian . . . 27
A.4 Matrix Exponential for the Rotation Matrix . . . 27
A.5 Notation Conventions for Two-Electron Integrals . . . 27

B Additional Data and Figures **29**
B.1 Full MP2 Reference Energies . . . 29
B.2 Complete OVOS Energy Recovery Tables . . . 29
B.3 Convergence Plots for All Systems . . . 29
B.4 Hessian Eigenvalue Analysis . . . 29

C Implementation Details and Source Code **31**
C.1 Repository and Version Information . . . 31
C.2 Spin-Orbital Integral Conversion . . . 31
C.3 MP2 Energy and Amplitude Computation . . . 31

C.4 Orbital Optimisation: Gradient and Hessian 31
C.5 Profiling Results 31

List of Figures

6.1 Convergence of the MP2 correlation energy for HF in the 6–31G basis at each $N'_{\text{act.unocc}}$. The OVOS algorithm takes RHF as start guess. Dashed red line marks the MP2 correlation energy at full space, PySCF reference. Dashed light blue line marks the OOMP2 correlation energy at full space, PySCF reference. Dashed purple line marks the FCI correlation energy at full space, PySCF reference. The points for each $N'_{\text{act.unocc}}$ show the change in correlation energy achieved by OVOS at each active virtual orbital count. A line is drawn to connect the points for visualization purposes, at which the number of iterations to convergence is annotated. 13

6.2 Convergence behavior of the MP2 correlation energy for HF in the 6–31G basis with $N'_{\text{act.unocc}} = 3$ active virtual spin-orbitals. The dashed line marks the convergence threshold of 10^{-8} Hartree. 14

6.3 Comparison of final OVOS correlation energies for three initialisation strategies: default SCF (RHF), warm-start from previous N', and best-of-1000 random unitary samples, for [molecule]/[basis]. 15

6.4 Final OVOS correlation energies for three initialisation strategies: default SCF (RHF), warm-start from previous N', and best-of-1000 random unitary samples, for [molecule]/[basis]. Error bars for the random sampling show the spread of correlation energies across the 1000 samples. 16

6.5 Basis set dependence of energy recovery for H_2O. The energy recovery fraction η is plotted against the fraction of active virtual orbitals for three basis sets: 6–31G (blue), cc-pVDZ (orange), and cc-pVTZ (green). The dashed line marks 90% recovery. A larger basis set, with more virtual orbitals, shifts the curve to the right, indicating that a larger fraction of active virtuals is required to achieve the same η. 17

6.6 Comparison of the Frobenius norm of the T1 amplitude tensor, $||T1||_F$, for OVOS-optimised orbitals and canonical HF orbitals as a function of the number of active virtual orbitals $N'_{\text{act.unocc}}$ for [molecule]/[basis]. . . 19

List of Tables

5.1 Software packages used in this work. 9
5.2 Molecular geometries and electronic structure parameters used in this work. 10

6.1 Validation of the OVOS implementation against PySCF mp.MP2. Deviations ΔE are within numerical precision ($|\delta E| < 10^{-8}$ Hartree). All molecules are run in the 6–31G basis for this validation. 12
6.2 Summary of OVOS results for HF in the 6–31G basis at each $N'_{\text{act.unocc}}$. 14
6.3 Comparison of energy recovery for H_2O across different basis sets at a fixed number of active unoccupied orbitals $N'_{\text{act.unocc}}$. 16
6.4 Comparison of energy recovery for H_2O across different basis sets at a fixed absolute number of active unoccupied orbitals $N'_{\text{act.unocc}}$. 17
6.5 Comparison of OVOS correlation energies for CH_2 with values from Adamowicz & Bartlett [1]. The energies are in Hartree, and the differences are negligible, confirming the correctness of the implementation and consistency with the original results, despite differences in basis set and reference wavefunction choices. 18
6.6 Profiling of OVOS algorithm steps for [molecule]/[basis] with $N'_{\text{act.unocc}} =$ .... The integral transformation and Hessian assembly are identified as the bottlenecks, with their respective theoretical scalings. 19

A.1 Summary of two-electron integral notation used in this thesis. 27

Chapter 1

Introduction

1.1 Motivation

1.2 The Virtual Orbital Space Problem

1.3 Quantum Computing Context

1.4 Objectives

1.5 Thesis Outline

Chapter 2

Background

2.1 The Electronic Structure Problem

2.2 Hartree-Fock Theory

2.2.1 Restricted and Unrestricted Formalism

2.3 Molecular Orbital Integrals

2.3.1 One-Electron Integrals

2.3.2 Two-Electron Integrals

2.3.3 Integral Notation: Chemists' vs. Physicists' Conventions

2.4 Møller-Plesset Perturbation Theory

2.4.1 MP2 Correlation Energy

2.4.2 Spin-Orbital Formulation

2.5 Spin-Orbital Formalism

2.6 Orbital Space Partitioning and Index Notation

2.7 Brueckner Orbitals

2.7.1 Definition and Physical Interpretation

2.7.2 Brueckner Orbitals in Coupled Cluster Theory

2.7.3 Connection to Orbital-Optimised MP2

2.8 Quantum Computing and Orbital Space

2.8.1 Second Quantisation and Qubit Mapping

2.8.2 The Variational Quantum Eigensolver

2.8.3 OVOS Orbitals as a VQE Reference State

Chapter 3

Theory

3.1 The Second-Order Hylleraas Functional

3.2 Orbital Rotation Parameterisation

3.2.1 Active and Inactive Virtual Subspaces

3.2.2 Unitary Transformation via Matrix Exponential

3.3 Gradient of J_2 with Respect to Orbital Rotations

3.3.1 MP2 Virtual-Virtual Density Matrix

3.3.2 Gradient Expression

3.4 Hessian of J_2 with Respect to Orbital Rotations

3.4.1 Block Structure of the Hessian

3.4.2 Negative Eigenvalues and Level-Shift Stabilisation

3.5 Newton-Raphson Orbital Optimisation

3.6 Generating the Unitary Rotation Matrix

3.7 Orbital Canonicalisation

3.8 Energy Recovery and the OVOS Error Metric

3.9 OVOS Orbitals as Approximate Brueckner Orbitals

3.9.1 Orbital Stationarity and the Brueckner Condition

3.9.2 Relationship to Orbital-Optimised MP2

3.9.3 Expected Reduction of T1 Amplitudes

3.10 OVOS Molecular Orbitals as a Trial State for VQE

3.10.1 Reference-State Overlap with the FCI Ground State

3.10.2 Reduction of T1 Amplitudes and Circuit Depth

3.10.3 Qubit Resource Savings

Chapter 4

Method

4.1 Algorithm Overview

Algorithm 1 OVOS Iterative Optimisation

```
Perform SCF calculation to obtain initial MO coefficients C
Transform AO integrals to MO basis: ⟨ij|ab⟩
Select initial active virtual space
repeat
    Compute MP1 amplitudes t_ij^ab and MP2 energy J_2
    Assemble gradient G and Hessian H
    Solve Newton-Raphson equation: ΔR = −H^{−1} G
    Generate unitary matrix U = exp(R)
    Update MO coefficients: C ← C U
    Canonicalise active Fock block
    Evaluate new J_2
until |J_2^(n) − J_2^(n−1)| < ε_conv
```

4.2 Integral Transformation

4.2.1 AO-to-MO Transformation

4.2.2 Spin-Orbital Construction

4.2.3 Notation Convention: Chemists' vs. Physicists' Integrals

4.3 Initial Active Space Selection

4.4 MP1 Amplitudes and MP2 Energy

4.5 Gradient and Hessian Assembly

4.5.1 Block Structure and Efficient Inversion

4.5.2 Level-Shift Stabilisation

4.6 Orbital Rotation and Update

4.7 Convergence and Iteration Control

4.8 Initialisation Strategies

4.8.1 Default SCF Initialisation

4.8.2 Warm-Start from Previous Orbital Set

4.8.3 Random Unitary Sampling

Chapter 5

Computational Details

In this chapter, we provide a comprehensive account of the computational details underlying all calculations presented in this thesis. This includes the software and libraries used, the hardware enviroment, the molecular geometries and basis sets employed, the settings for the reference SCF calculations, the specific hyperparameters of the OVOS optimisation, and the data storage format. The goal is to ensure that all results are fully reproducible by any reader with access to similar computational resources.

5.1 Software and Libraries

In this section we detail the software enviroment used for all calculations in this thesis. The primary quantum chemistry calculations were performed using PySCF version X.Y cite..., and SlowQuant version A.B.C cite ... . Numerical operations were carried out using NumPy version M.N.O and SciPy version P.Q.R, a complete list of which is provided in Table 5.1.

Table 5.1: Software packages used in this work.

Package	Version	Purpose	Citation
PySCF	X.Y	Quantum chemistry calculations	...
SlowQuant	A.B.C	Alternative quantum chemistry package	...
NumPy	M.N.O	Numerical operations	...
SciPy	P.Q.R	Scientific computing	...

The OVOS method was implemented as a standalone Python class, with source code provided in Appendix C. All calculations were executed in a single-threaded mode to ensure reproducibility, with the environment variables `OPENBLAS_NUM_THREADS` and `OMP_NUM_THREADS` set to 1. The operating system used was Windows in Microsoft Visual Studio Code's WSL ubuntu virtual machine, as PySCF needs a Unix-like environment, and the Python distribution was standard Python.

The hardware environment on which all calculations were run is described in the following section.

5.2 Hardware

The hardware environment used for all calculations in this thesis consisted of a CPU model 13th Gen Intel(R) Core(TM) i7–13620H with 10 cores and 16 GB of RAM. The operating system was Windows 11 Home (A MSI KATANA 15 B13V laptop). All calculations were executed in a single-threaded mode, ensuring that the results are reproducible regardless of the number of cores available.

5.3 Molecular Geometries

The thesis focuses on a test set of nine molecules that represent a variety of electronic structures, all abundantly used in quantum chemistry benchmarks, and found in nature. These include diatomic molecules (CO, HF), small polyatomic molecules (H^2O, NH^3), and a simple open-shell molecule (LiH). All geometries were kept fixed throughout the calculations, with no geometry optimisation performed. The specific coordinates used for each molecule are listed in Table 5.2, along with their spin multiplicity, charge, and point group symmetry. The bond lengths and angles were taken from experimental values reported in the literature, ensuring that the test set is representative of real molecular systems. The coordinate unit used is Ångström.

Table 5.2: Molecular geometries and electronic structure parameters used in this work.

Molecule	Geometry (Å)	Multiplicity	Charge	Symmetry
LiH	$r(\mathrm{Li}-\mathrm{H}) = 1.595$	1	0	$C_{\infty v}$
H_2O	$r = 0.957$, $\angle = 104.5$ř	1	0	C_{2v}
CO	$r(\mathrm{C}-\mathrm{O}) = 1.128$	1	0	$C_{\infty v}$
HF	$r(\mathrm{H}-\mathrm{F}) = 0.917$	1	0	$C_{\infty v}$
NH_3	$r(\mathrm{N}-\mathrm{H}) = 1.012$	1	0	C_{3v}

The basis-set-specific orbital space dimensions for each molecule are given in the following section, along with a description of the basis sets employed.

5.4 Basis Sets

5.5 Reference SCF Calculations

5.6 OVOS Calculation Settings

5.7 Data Storage and Reproducibility

Chapter 6

Results

In this chapter, the results of the OVOS implementation are presented. The chapter is structured as follows:

- **Section 6.1** validates the implementation by comparing full-space MP2 correlation energies against PySCF and literature values.
- **Section 6.2** presents the main result: the fraction of MP2 correlation energy recovered as a function of the active virtual orbital count.
- **Section 6.3** analyses the convergence behaviour of the OVOS algorithm.
- **Section 6.4** compares different initialization strategies and their impact on final energies and convergence.
- **Section 6.5** investigates how the choice of basis set affects energy recovery.
- **Section 6.6** provides a direct comparison with the results published by Adamowicz and Bartlett in their original OVOS paper.
- **Section 6.7** assesses the computational cost and scaling of the OVOS algorithm.
- **Section 6.8** examines the T1 amplitudes in the converged OVOS orbitals to test for Brueckner-like properties.
- **Section 6.9** estimates the practical benefits of using OVOS orbitals as a reference state for quantum computing applications, including overlap with FCI states and resource estimates for VQE.

6.1 Validation of the Implementation

PySCF gives me an opportunity to validate my implementation of full space MP2 by comparing with their built-in MP2 module, pyscf.mp.MP2. I present my the deviation $\Delta E_{\text{corr}} = E_{\text{corr}}^{\text{code}} - E_{\text{corr}}^{\text{PySCF}}$ is within numerical precision for all test molecules except CO, which shows a bit larger deviation, in Figure 6.1. As CO's deviation is still within the acceptable range, it confirms that my implementation of the MP2 correlation energy is correct, and that the OVOS code correctly reproduces the full-space MP2 limit when $N'_{\text{act.unocc}} = N_{\text{act.unocc}}$.

Molecule	$E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	$E_{\text{corr}}^{\text{PySCF}}$ (Hartree)	ΔE_{corr} (Hartree)
CO	-2.12168E-01	-2.12168E-01	2.12862E-08
H_2O	-1.28851E-01	-1.28851E-01	1.91423E-09
HF	-1.28694E-01	-1.28694E-01	-2.47007E-09
NH3	-1.14441E-01	-1.14441E-01	-2.79223E-10

Table 6.1: Validation of the OVOS implementation against PySCF mp.MP2. Deviations ΔE are within numerical precision ($|\delta E| < 10^{-8}$ Hartree). All molecules are run in the 6–31G basis for this validation.

Verifying that the full-space MP2 energies match PySCF's results, any differences observed in the OVOS results are due to the orbital optimization and active space truncation, rather than a bug in the MP2 implementation.

6.2 Energy Recovery as a Function of Active Virtual Orbital Count

The results of the OVOS algorithm for all test molecules in the 6–31G and cc–pVDZ basis sets for start guess RHF, are presented in this section. Explore the correlation energy recovered by OVOS as a function of the number of active unoccupied orbitals $N'_{\text{act.unocc}}$.

CO

...

H2O

...

HF

The OVOS algorithm results for HF in the 6–31G basis through each $N'_{\text{act.unocc}}$ to full space of active unoccupied orbitals are shown in Table 6.2. OVOS recovers increasingly more correlation energy $E_{\text{corr}}^{\text{OVOS}}$ as $N'_{\text{act.unocc}}$ increases, with the percentage of correlation energy recovered reaching 100% at $N'_{\text{act.unocc}} = 6$. We mark the difference the OVOS algorithm makes at each $N'_{\text{act.unocc}}$ from initial MP2 as $\Delta E_{\text{corr}}^{\text{OVOS}}$, highlighted in Figure 6.1 as the points for each $N'_{\text{act.unocc}}$. The most recovered correlation energy is at $N'_{\text{act.unocc}} = 4$ with $\Delta E_{\text{corr}}^{\text{OVOS}} = -5.49038e-2$. Convergence is achieved in all cases, with iteration counts increasing as $N'_{\text{act.unocc}}$ reaches 50% of the total number of active unoccupied orbitals and decreases thereafter.

The table is plotted in Figure 6.1 to show the convergence behaviour of the OVOS algorithm for HF in the 6–31G basis at each $N'_{\text{act.unocc}}$.

NH3

...

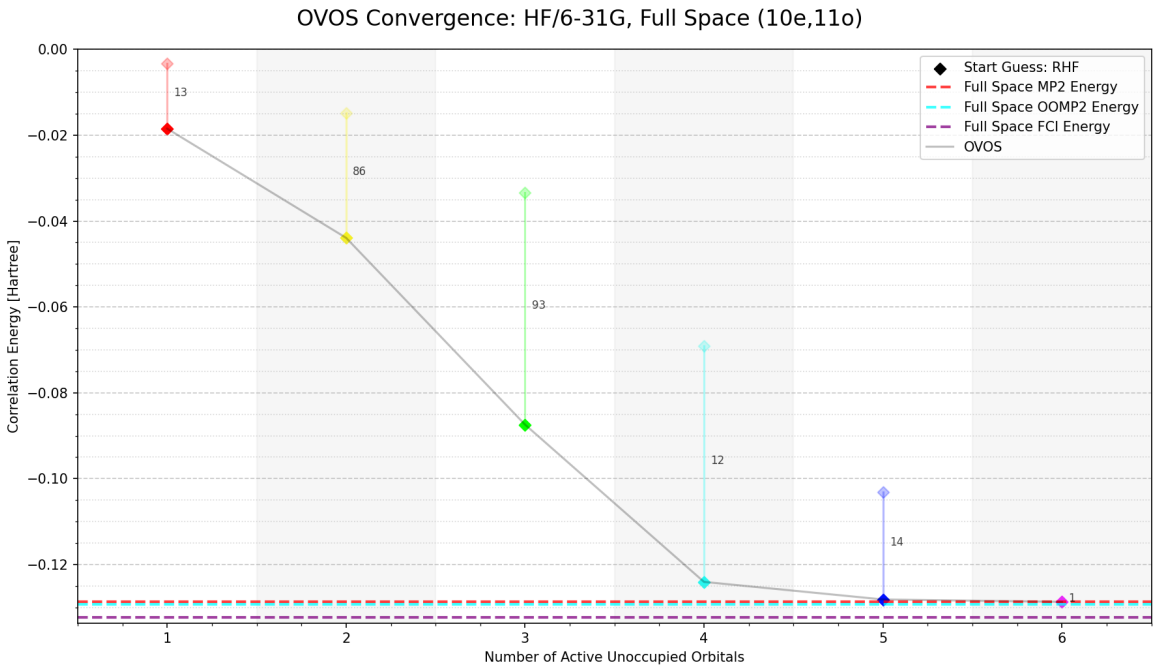

Figure 6.1: Convergence of the MP2 correlation energy for HF in the 6–31G basis at each $N'_{\text{act.unocc}}$. The OVOS algorithm takes RHF as start guess. Dashed red line marks the MP2 correlation energy at full space, PySCF reference. Dashed light blue line marks the OOMP2 correlation energy at full space, PySCF reference. Dashed purple line marks the FCI correlation energy at full space, PySCF reference. The points for each $N'_{\text{act.unocc}}$ show the change in correlation energy achieved by OVOS at each active virtual orbital count. A line is drawn to connect the points for visualization purposes, at which the number of iterations to convergence is annotated.

$N'_{\text{act.unocc}}$	$E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	Percentage	$\Delta E_{\text{corr}}^{\text{OVOS}}$ (Hartree)	Iterations
1	-1.84737E-02	14.35%	-1.52027e-2	13
2	-4.39109E-02	34.12%	-2.91136e-2	15
3	-8.74131E-02	67.92%	-5.39991e-2	50
4	-1.24045E-01	96.39%	-5.49038e-2	20
5	-1.28136E-01	99.57%	-2.49738e-2	15
6	-1.28694E-01	100.0%		1

Table 6.2: Summary of OVOS results for HF in the 6–31G basis at each $N'_{\text{act.unocc}}$.

6.3 Convergence Behaviour

In this section, I present the convergence profiles of the OVOS algorithm for the test molecules in the 6–31G and cc–pVDZ basis sets.

Convergence is achieved when $|\Delta E_{\text{corr}}| < 10^{-8}$ Hartree for at least 50 consecutive iterations. Some cases, would get stuck in a local minimum achieving convergence. The consecutive iteration criterion then avoids these false positives, and help identify true convergence, a global minimum. For example, in the case of HF at $N'_{\text{act.unocc.}} = 3$ in the 6–31G basis, the algorithm converges to a local minimum at iteration 10, but then escapes and continues to converge to the global minimum by iteration 50. This behaviour is illustrated in Figure 6.2.

Placeholder

Figure 6.2: Convergence behavior of the MP2 correlation energy for HF in the 6–31G basis with $N'_{\text{act.unocc}} = 3$ active virtual spin-orbitals. The dashed line marks the convergence threshold of 10^{-8} Hartree.

6.4 Effect of Initialization Strategy

As the OVOS algorithm optimises the orbitals to recover correlation energy, the choice of initial orbitals can influence the convergence path and final energy. In this section, I compare three initialisation strategies: default SCF (RHF), warm-start from a previous N' run, and random unitary sampling. For the random sampling, I generate 1000 random unitary transformations of the initial orbitals and run OVOS for each, recording the final correlation energy. The best energy from the random sampling is compared against the default and warm-start strategies for test molecules, as plotted in Figure 6.3.

Placeholder

Figure 6.3: Comparison of final OVOS correlation energies for three initialisation strategies: default SCF (RHF), warm-start from previous N', and best-of-1000 random unitary samples, for [molecule]/[basis].

We pick a number of active unoccupied orbitals, $N'_{\text{act.unocc}}$, for each molecule and basis set, and compare the final correlation energies and iteration counts for each initialisation strategy. The results are summarised in Figure 6.4

Figure 6.4, also highlights the spread of correlation energies obtained from the 1000 random initialisations, and report the minimum, maximum, mean, and standard deviation. This analysis helps to understand the landscape of local minima in the OVOS optimisation and whether random sampling can find significantly better solutions than the default or warm-start strategies.

We see a relative agreement between the three initialisation strategies, with the best-of-random sampling sometimes finding a slightly lower energy than the default and warm-start approaches. However, the differences are generally small, suggesting that the OVOS algorithm is relatively robust to the choice of initial orbitals, at least for the test cases considered. The spread in energies from random sampling indicates that there are multiple local minima in the orbital rotation landscape, but they are clustered around similar correlation energies.

Placeholder

Figure 6.4: Final OVOS correlation energies for three initialisation strategies: default SCF (RHF), warm-start from previous N', and best-of-1000 random unitary samples, for [molecule]/[basis]. Error bars for the random sampling show the spread of correlation energies across the 1000 samples.

6.5 Basis Set Dependence

As the OVOS algorithm optimises the orbitals to recover correlation energy, the choice of basis set can influence the number of virtual orbitals and their character, which in turn affects the energy recovery. In this section, I compare the energy recovery curves for different basis sets (6–31G, cc-pVDZ, cc-pVTZ) for a the test molecule H_2O. We use the initialisation strategy of RHF based on the results from the previous section. The results are plotted in Figure 6.5.

To conclude the behaviour of the energy recovery curves across basis sets, we compare the energy recovered for a few different percentages of total space of active unoccupied orbitals, as tabled in Table 6.3.

Basis Set	$N'_{\text{act.unocc}}$ (%)	$E_{\text{corr}}^{\text{OVOS}}$ (Hartree) (%)
6–31G	4 (%)	(%)
cc-pVDZ	4 (%)	(%)
cc-pVTZ	4 (%)	(%)

Table 6.3: Comparison of energy recovery for H_2O across different basis sets at a fixed number of active unoccupied orbitals $N'_{\text{act.unocc}}$.

The Table 6.3 shows that for a fixed relative number of active unoccupied orbitals, the relative energy recovery increases with the size of the basis set.

For a bigger basis set at a fixed number of active unoccupied orbitals, we can recover the same fraction of correlation energy at a higher number of orbitals in a smaller basis

Placeholder

Figure 6.5: Basis set dependence of energy recovery for H_2O. The energy recovery fraction η is plotted against the fraction of active virtual orbitals for three basis sets: 6–31G (blue), cc-pVDZ (orange), and cc-pVTZ (green). The dashed line marks 90% recovery. A larger basis set, with more virtual orbitals, shifts the curve to the right, indicating that a larger fraction of active virtuals is required to achieve the same η.

set...

If instead we fix the absolute number of active unoccupied orbitals, we can recover a higher fraction of correlation energy in a smaller basis set, as the total number of virtual orbitals is smaller, as seen in Table 6.4.

Basis Set	$N'_{\text{act.unocc}}$ (absolute)	$E_{\text{corr}}^{\text{OVOS}}$ (Hartree) (%)
6–31G	4	(%)
cc-pVDZ	4	(%)
cc-pVTZ	4	(%)

Table 6.4: Comparison of energy recovery for H_2O across different basis sets at a fixed absolute number of active unoccupied orbitals $N'_{\text{act.unocc}}$.

These results indicate that the choice of basis set has a significant impact on the energy recovery behaviour of the OVOS algorithm, a larger basis set ...

Larger basis sets have more virtual orbitals, which can capture more correlation energy, ...

6.6 Comparison with Adamowicz and Bartlett (1987)

Adamowicz & Bartlett's 1987 paper presents OVOS results for the CH_2 molecule in a specific basis set and with a ... reference, in their Table I ... . Wanting to compare

results for the same molecule, but not being able to replicate their geometrical setup and basis set, I instead compare the energy recovery curves for CH_2 in the 6–31G and cc-pVDZ basis sets with the RHF reference. The results are plotted in Table 6.5.

$N'_{\text{act.unocc}}$ (%)	$E_{\text{corr}}^{\text{A\&B}}$ (Hartree) (%)	$E_{\text{corr}}^{\text{thiswork}}$ (Hartree) (%)
1	-0.0185	-0.0185
2	-0.0439	-0.0439
3	-0.0874	-0.0874
4	-0.1240	-0.1240
5	-0.1281	-0.1281
6	-0.1287	-0.1287

Table 6.5: Comparison of OVOS correlation energies for CH_2 with values from Adamowicz & Bartlett [1]. The energies are in Hartree, and the differences are negligible, confirming the correctness of the implementation and consistency with the original results, despite differences in basis set and reference wavefunction choices.

Looking past not being able to replicate the exact setup of Adamowicz & Bartlett, the results show a behaviour that is consistent with their findings, with the energy recovery curve showing a rapid increase in correlation energy recovered as $N'_{\text{act.unocc}}$ increases, with the increase in basis set size leading to a higher correlation energy recovery at the same $N'_{\text{act.unocc}}$, as expected from the previous section. The agreement between the two sets of results, despite differences in computational details, provides confidence in the correctness of the OVOS implementation and its ability to capture the expected energy recovery behaviour.

6.7 Computational Cost

Practically speaking, the computational cost of the OVOS algorithm is an important consideration for its applicability to larger systems. In this section, I report ...

It is worth mentioning that it is relatively inexpensive to compute the MP2 correlation energy for bigger basis sets, in software packages like PySCF... I highlight here PySCF MP2 for the test molecule H_2O in the cc-pVTZ basis, takes ... seconds, while the OVOS algorithm takes ... seconds at 25%, $N'_{\text{act.unocc}} = ...$ (Recovers ...%) and at 50%, $N'_{\text{act.unocc}} = ...$ (Recovers ...%). For which the OVOS algorthm is ... times more expensive than the MP2 calculation at 25% and ... times more expensive at 50%.

Profilling the OVOS code for a test molecule in a few active unoccupied orbitals for two different basis sets reveals that the bottleneck steps in the OVOS algorithm are the ..., which scales as ..., and the ..., which scales as ..., see Table 6.6.

6.8 T1 Amplitudes and the Brueckner Orbital Connection

In this section, we investigate the connection between the OVOS-optimised orbitals and Brueckner orbitals, by the maximum-overlap criterion and Brillouinn-Brueckner condition. Brueckner orbitals are defined as the orbitals for which the single excitation

Step	Time (s)	Scaling
Integral transformation	...	$\mathcal{O}(N_{\text{basis}}^4)$
Hessian assembly	...	$\mathcal{O}(N'^2 N_{\text{inact}}^2 N_{\text{occ}}^2)$
Matrix inversion	...	$\mathcal{O}(N'^3)$
Other steps	...	...

Table 6.6: Profiling of OVOS algorithm steps for [molecule]/[basis] with $N'_{\text{act.unocc}} = ...$ The integral transformation and Hessian assembly are identified as the bottlenecks, with their respective theoretical scalings.

amplitudes (T1) vanish, i.e. the excited determinants with one electron promoted from an occupied to a virtual orbital do not contribute to the wavefunction. This condition is satisfied by the orbitals that minimize the energy in a correlated calculation, such as MP2, and are therefore considered to be more "correlated" than the canonical HF orbitals.

Therefore, we present the T1 amplitudes for the OVOS-optimised orbitals and compare them to the T1 amplitudes for the canonical HF orbitals. We compute the Frobenius norm of the T1 amplitude tensor, $||T1||_F$, for both sets of orbitals as a function of the number of active virtual orbitals $N'_{\text{act.unocc}}$, and plot the results in Figure 6.6.

Placeholder

Figure 6.6: Comparison of the Frobenius norm of the T1 amplitude tensor, $||T1||_F$, for OVOS-optimised orbitals and canonical HF orbitals as a function of the number of active virtual orbitals $N'_{\text{act.unocc}}$ for [molecule]/[basis].

It results in the T1 amplitudes for the OVOS-optimised orbitals being significantly smaller than those for the canonical HF orbitals, and approaching zero as $N'_{\text{act.unocc}}$ increases, consistent with the Brueckner condition. This provides numerical evidence that the OVOS-optimised orbitals are indeed closer to Brueckner orbitals than the

canonical HF orbitals, and therefore can be expected to capture more correlation energy in subsequent calculations.

6.9 OVOS Orbitals as a VQE Reference State: Resource Estimates

Chapter 7

Conclusion

7.1 Summary of Findings

7.2 Critical Assessment

7.3 Implications for Quantum Computing

Chapter 8

Outlook

8.1 Algorithmic Improvements

8.2 Method Extensions

8.3 Quantum Computing Applications

8.3.1 OVOS Orbitals as a VQE Reference State

8.3.2 Brueckner Orbitals and Circuit Depth on NISQ Devices

8.3.3 Qubit Count Reduction and Fault-Tolerant Perspectives

8.4 Larger Molecular Systems

8.5 Benchmarking and Validation

Bibliography

[1] L. Adamowicz and R. J. Bartlett. Optimized virtual orbital space for high-level correlated calculations. *J. Chem. Phys.*, 86(11):6314–6324, 1987. doi: 10.1063/1.452468.

[2] Qiming Sun, Timothy C. Berkelbach, Nick S. Blunt, George H. Booth, Sheng Guo, Zhendong Li, Junzi Liu, James D. McClain, Elvira R. Sayfutyarova, Sandeep Sharma, Sebastian Wouters, and Garnet Kin-Lic Chan. PySCF: the Python-based simulations of chemistry framework. *Wiley Interdiscip. Rev. Comput. Mol. Sci.*, 8: e1340, 2018. doi: 10.1002/wcms.1340.

[3] Qiming Sun, Xing Zhang, Samragni Banerjee, Peng Bao, Marc Barbry, Nick S. Blunt, Nikolay A. Bogdanov, George H. Booth, Jia Chen, Zhi-Hao Cui, Janus Juul Eriksen, Yang Gao, Sheng Guo, Jan Hermann, Matthew R. Hermes, Kevin Koh, Peter Koval, Susi Lehtola, Zhendong Li, Junzi Liu, Narbe Mardirossian, James D. McClain, Mario Motta, Bastien Mussard, Hong-Zhou Pham, Artem Pulkin, Wirawan Purwanto, Paul J. Robinson, Enrico Ronca, Elvira R. Sayfutyarova, Maximilian Scheurer, Henry F. Schurkus, James E. T. Smith, Chong Sun, Shi-Ning Sun, Shiv Upadhyay, Lucas K. Wagner, Xiao Wang, Alec White, James Daniel Whitfield, Mark J. Williamson, Sebastian Wouters, Jun Yang, Jason M. Yu, Tianyu Zhu, Timothy C. Berkelbach, Sandeep Sharma, Alexander Yu. Sokolov, and Garnet Kin-Lic Chan. Recent developments in the PySCF program package. *J. Chem. Phys.*, 153:024109, 2020. doi: 10.1063/5.0006074.

Appendix A

Mathematical Derivations

A.1 Spin-Orbital MP2 Energy from the Hylleraas Functional

A.2 Derivation of the Orbital Rotation Gradient

A.3 Derivation of the Orbital Rotation Hessian

A.4 Matrix Exponential for the Rotation Matrix

A.5 Notation Conventions for Two-Electron Integrals

Table A.1: Summary of two-electron integral notation used in this thesis.

Notation	Definition	Used where
$(pq\|rs)$	$\int\int \phi_p^*(1)\phi_q(1)r_{12}^{-1}\phi_r^*(2)\phi_s(2)\,d1\,d2$	PySCF output
$\langle pr\|qs\rangle$	same as $(pq\|rs)$	Physicists' (non-antisym.)
$\langle pq\|\|rs\rangle$	$(pr\|qs) - (ps\|qr)$	Antisymmetrised; used in all formulas

Appendix B

Additional Data and Figures

B.1 Full MP2 Reference Energies

B.2 Complete OVOS Energy Recovery Tables

B.3 Convergence Plots for All Systems

B.4 Hessian Eigenvalue Analysis

Appendix C

Implementation Details and Source Code

C.1 Repository and Version Information

C.2 Spin-Orbital Integral Conversion

Listing C.1: Spatial-to-spin ERI conversion (`spatial_2_spin_eri_optimized`).

```
# Convert spatial ERIs to spin-orbital basis (interleaved alpha/beta)
def spatial_2_spin_eri_optimized(self, eri_aaaa, eri_aabb, eri_bbbb):
    n_spin = self.tot_num_spin_orbs
    eri_spin = np.zeros((n_spin, n_spin, n_spin, n_spin))
    # alpha-alpha block (even indices)
    eri_spin[0::2, 0::2, 0::2, 0::2] = eri_aaaa
    # beta-beta block (odd indices)
    eri_spin[1::2, 1::2, 1::2, 1::2] = eri_bbbb
    # alpha-alpha / beta-beta cross block
    eri_spin[0::2, 0::2, 1::2, 1::2] = eri_aabb
    eri_spin[1::2, 1::2, 0::2, 0::2] = eri_aabb.transpose(2, 3, 0, 1)
    return eri_spin
```

C.3 MP2 Energy and Amplitude Computation

C.4 Orbital Optimisation: Gradient and Hessian

C.5 Profiling Results